

CS 435 - 472: Machine Learning Project Guidelines

Project Introduction

The machine learning (ML) project is an opportunity to apply the concepts and methods studied in class by addressing a real-world problem using ML. This project, to be completed in groups of up to **3 students**, aims to develop a comprehensive understanding of the ML project pipeline, from data collection and preparation to model evaluation.

Important Date:

- **May 3, 2026:** Project submission deadline (report, code, and presentation) via the Blackboard platform.

1. Project Report Content

a) Project Title

- Each project should have a clear title summarizing the project's objective.

b) Project Overview

- **Objective:** Briefly describe the objective of the project and the problem you aim to solve using ML.
- **Expected Outcomes:** Specify the results you hope to achieve. This section serves to contextualize the significance of your work.

c) Dataset

- **Dataset Source:** Use publicly available datasets, such as those from Kaggle or the UCI Machine Learning Repository.
- **Description:** Briefly describe the content of your dataset (size, variables, data types).
- **Data Preparation:** Explore and clean your dataset. Describe the preprocessing techniques applied (handling missing values, normalization, etc.) and visualize relevant descriptive statistics.

d) Model Selection and Training

- **Model Choice:** Select one or more ML algorithms covered in class (and possibly other relevant methods).
- **Training:** Apply these models on the training dataset after splitting the data into training and test sets.

e) Model Evaluation and Testing

- **Evaluation:** Test your model on the test dataset and use appropriate metrics (accuracy, recall, F1-score, etc.) to evaluate its performance.
- **Result Analysis:** Interpret the results obtained, identifying the model's strengths and limitations.

f) Discussion

- **Key Findings:** Analyze the main results and what they reveal about the model and the problem studied.

g) Conclusion

- **Summary:** Conclude by summarizing the project, the challenges encountered, the lessons learned, and the impact of the model on the problem addressed.

h) Optional / Advanced Tasks

To further improve your project, you may consider some of the following:

- Compare **different ML families** (e.g., Decision Tree vs. Neural Network,...) and discuss trade-offs.
- Integrate **visual explanations** (plots, heatmaps, or confusion matrix).
- Implement a **Deep Learning approach** (e.g., using TensorFlow or PyTorch) and compare its performance with classical machine learning models.
- Develop a simple **graphical user interface (GUI)** that connects to your trained model (backend) to allow users to input data and obtain predictions.
- *(Note: These tasks are optional but can positively impact your project evaluation and demonstrate deeper exploration of machine learning techniques.)*

2. Project Presentation

Prepare a **10-minute presentation** to summarize your project. The presentation should be divided equitably among group members to ensure fair participation.

Slide Content:

1. Title and team members.
2. Introduction and project objective.
3. Dataset and preparation methods.
4. Selected model and justification.
5. Training results and model performance.
6. Discussion and conclusions.
7. Future perspectives and potential improvements.
8. Questions and acknowledgments.

3. Project Code

Your code should be clear, well-structured, and appropriately commented. Key functions and crucial steps should be explained in comments to ensure ease of understanding.

4. Submission on Blackboard

- **Documents to submit:** Your code, the project report, and the presentation.
- **Submission format:** Only one group member should submit all files, ensuring that each file includes the names and IDs of all group members.

Notice: The choice, originality, creativity, and quality of the idea and the example will directly affect the grade. Projects that simply replicate existing online notebooks without interpretation will receive limited credit.